Credit Risk Analysis and Scoring Model

1. Executive Summary

The aim of the following project is to develop a model capable of identifying and classifying customers with high credit risk using demographic and financial information.

Credit risk refers to the possibility that a counterparty, such as a customer or supplier, may fail to meet its payment obligations towards an organization. It can manifest in different ways, ranging from delayed payments to complete default. Credit risk management is fundamental for a company's financial stability, as it represents one of the main threats to business continuity.

In order to identify customers with high credit risk, a logistic regression model was estimated. The choice of this model was driven not only by its predictive capabilities but also by its interpretability compared to black-box models.

Two main tools were used for this project: Python for the modeling phase and Power BI for the creation of a dashboard aimed at visualizing the obtained results.

The final model achieved a good AUC value equal to 0.80, showing an improvement compared to the initial model. Improvements were also observed in the main evaluation metrics, including Precision, Recall, and F1-score. The final predicted probabilities were then used to build a credit scoring system through which customers were divided into three risk categories: low, medium, and high risk.

The usefulness of the model goes beyond estimating the probability that a customer will be reliable or risky. It can support decision-making processes in credit approval, help reduce economic losses related to defaults, and allow faster and more standardized customer evaluations. Furthermore, the Low/Medium/High Risk segmentation facilitates monitoring prioritization for riskier customers and can be integrated into dashboards and monitoring systems.

2. Dataset Description

The data used in this project comes from the UCI Machine Learning Repository and is known as the German Credit Dataset. The dataset is publicly available and widely used for credit scoring and binary classification problems.

The dataset is composed of 20 independent variables and one target variable, for a total of 1000 observations. Both numerical and categorical variables are included, covering financial as well as personal characteristics of customers.

The binary target variable is credit_risk, which represents customer creditworthiness divided into good credit and bad credit customers, distributed approximately as 70% good and 30% bad. The main explanatory variables are reported in Table 1.

Variables	Description	Utility
amount	Amount of credit requested	Possible indicator of financial exposure
duration	Loan duration	Longer loans may imply higher risk
savings	Customer savings availability	Possible indicator of economic stability
status	Checking account status	Useful for evaluating liquidity and financial reliability
age	Customer age	Possible relationship with economic and employment stability
employment_duration	Employment duration	Indicator of income continuity

Table 1- Description of the Explanatory Variables

The set of variables allows the study of the phenomenon; however, the dataset is not free from limitations. First, the number of observations is relatively small (1000), which may limit the accuracy and stability of the estimated model. Furthermore, advanced financial information is missing, potentially leading to the exclusion of important factors influencing credit risk. In addition, since the data is not recent, the model may have limited generalizability to modern banking contexts.

On the other hand, one strength of the dataset is that the variables are relatively easy to read and interpret. This can be useful for observing social and demographic aspects influencing credit risk without relying exclusively on highly technical financial indicators.

Another, although less critical, limitation is the class imbalance in the target variable. As previously mentioned, the dataset contains an approximate 70/30 proportion in favor of the Good class. Although a balanced distribution between classes would generally be preferable for model estimation, this imbalance was not considered severe enough to require additional balancing techniques.

3. Data Preparation and Preprocessing

3.1 Descriptive Analysis

As a first step in the analysis, an exploratory study of the main variables in the dataset was conducted. It was observed that the average loan amount was 2985 Deutsche Marks for good customers and 3938 Deutsche Marks for bad customers, corresponding to a difference of approximately 900 euros when adjusted for inflation. Loan duration also differed between the two groups: high-risk customers requested loans with an average duration approximately 5 months longer than low-risk customers (24 months versus 19 months). No substantial differences were observed in average age, which ranged from 33 years for bad customers to 36 years for good customers.

Considering the loan amount distribution without separating customers by risk category, Graph 1 shows a strongly right-skewed distribution, with most loans below 5000 Deutsche Marks and a long tail extending up to approximately 17500 marks. This distribution highlights one of the dataset's limitations, namely its focus on relatively small loans while excluding larger and potentially riskier credit exposures, such as mortgage loans.

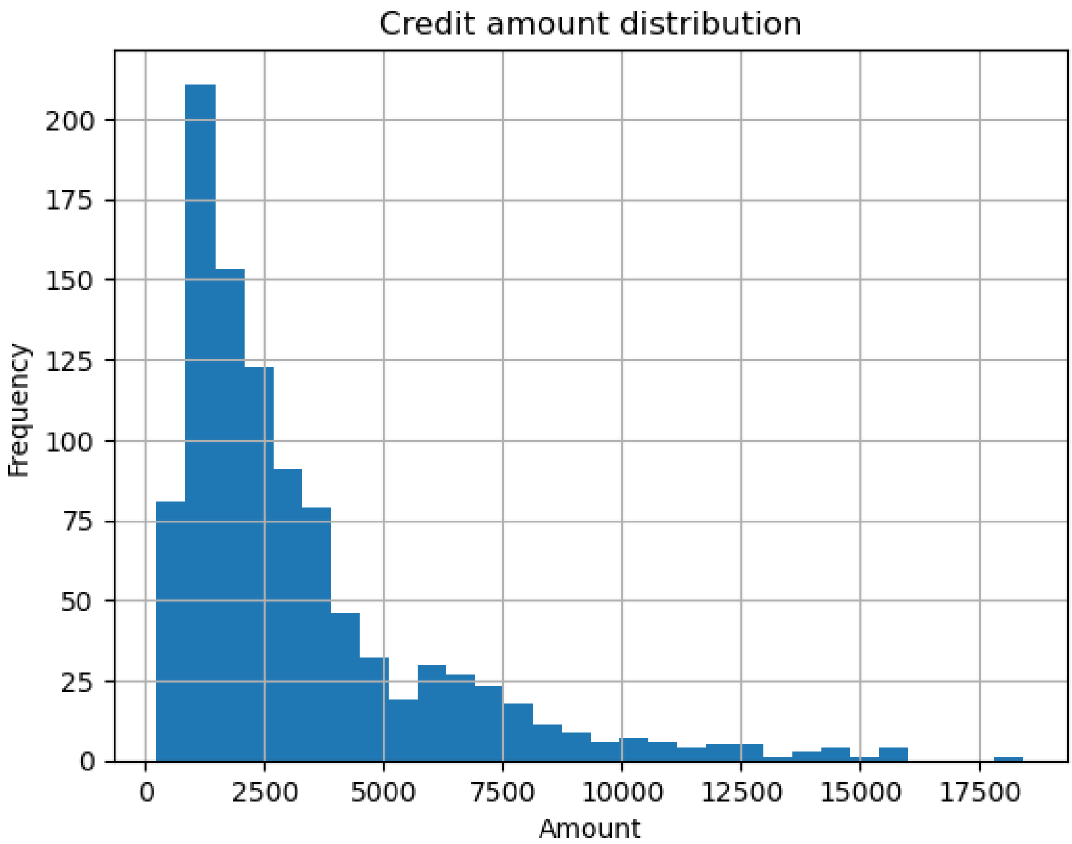


Figure 1 – Frequency Distribution of Granted Loan Amounts

Analyzing the categorical variables revealed an interesting pattern regarding savings accounts, with increasingly larger differences in favor of low-risk customers as savings increased. In the highest savings category (savings above 1000 Deutsche Marks), the distribution consisted of 87.5% good customers and only 12.5% bad customers. A similar, although less pronounced, pattern was also observed for checking account status categories. Further confirmation that lower-risk customers generally possess a stronger financial base can be found in the housing variable, where a larger gap was observed among customers owning their homes.

3.2 Preprocessing

Before estimating the model, several checks on the dataset and potential data-related issues were performed. First, missing values were analyzed, and fortunately no missing data problems were identified.

Subsequently, due to the presence of several categorical variables, one-hot encoding was applied. This technique replaces a categorical factor variable with multiple binary variables indicating the presence or absence of each category. The advantages of this approach include the removal of artificial ordinality in categorical variables, a more detailed representation of the information, improved model performance, and compatibility with algorithms requiring numerical inputs only.

As a final step before model estimation, the dataset was divided into training and testing subsets. This split was necessary in order to validate the model and evaluate its predictive reliability. A stratified split was used to preserve the original class distribution between good and bad customers.

One issue related to the train-test split concerns the relatively small number of observations. Reserving part of the dataset for testing may negatively affect model estimation. As an alternative, an internal validation technique such as cross-validation could have been used.

After estimating the initial model, it was used to investigate multicollinearity and outliers. Multicollinearity was assessed through the Variance Inflation Factor (VIF), and the highest values were associated with the variables foreign_worker_yes, purpose_domestic appliances, and other_debtors_none. These three variables were removed from the model. Although additional variables also presented relatively high VIF values, they were retained due to their potential importance for the final model and because the overall VIF values decreased after removing the three most problematic variables.

Outliers were identified using the Interquartile Range (IQR) method, leading to the exclusion of 138 observations.

4. Model Development and Evaluation

4.1 Model Estimation and Validation

After addressing multicollinearity and outlier-related issues, the final model was estimated. As previously discussed, logistic regression was selected because it provides good predictive performance while also offering a level of interpretability that many other models do not provide. The presence of coefficients and p-values for each variable makes it possible not only to classify customers but also to investigate which variables are statistically significant and how they influence the final predicted probability.

After training the model on the training dataset, several evaluation metrics were computed using the test dataset. The first metric considered was the AUC, an excellent indicator of model quality and discriminative power. The final result was an AUC equal to 0.80, indicating good predictive capability and overall model reliability. This value was achieved partly thanks to the preprocessing operations previously performed, which increased the AUC from approximately 0.75 to 0.80.

Before evaluating the final classification metrics, threshold tuning was performed. The default threshold of 0.5 was adjusted in order to improve classification performance. The behavior of the metrics as the threshold changes is illustrated in Graph 2.

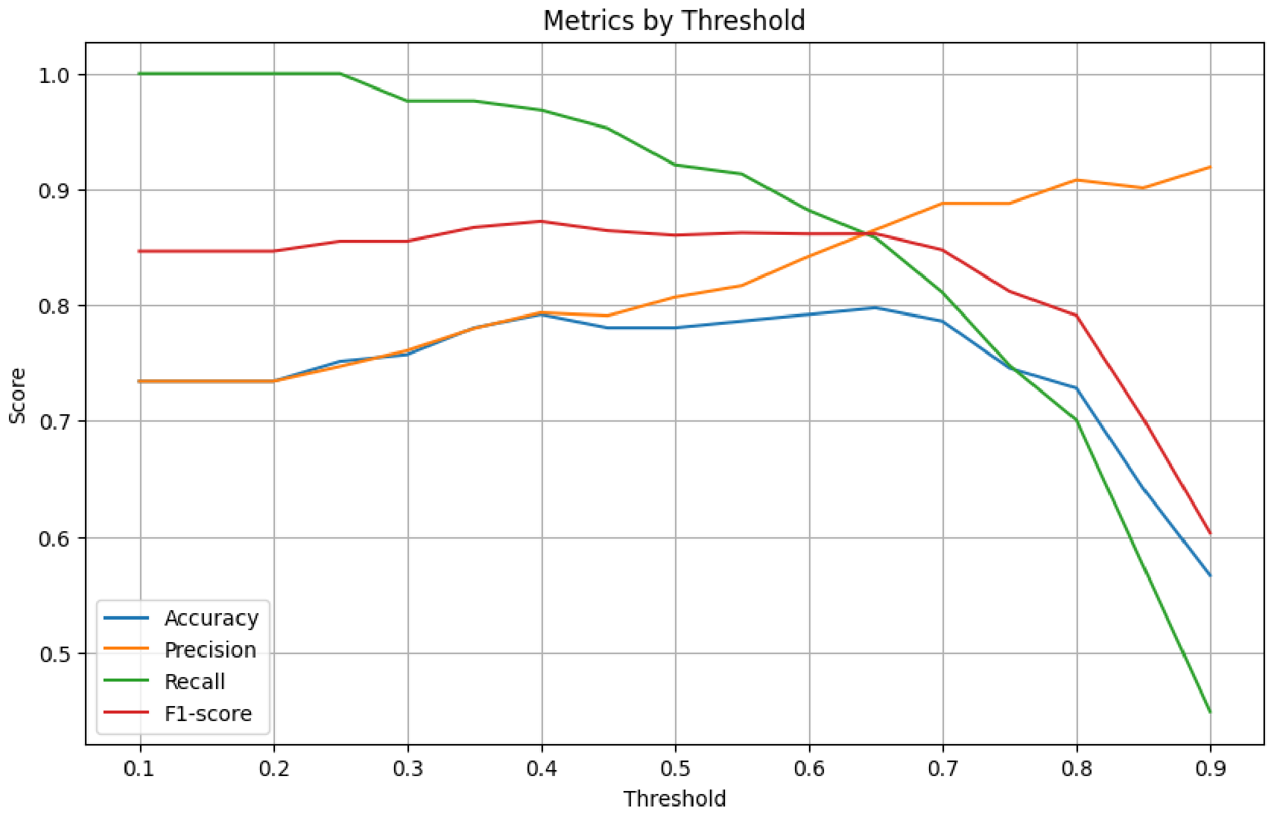


Figure 2 – Performance Metrics Across Different Classification Thresholds

From Graph 2, it can be observed that the optimal threshold lies between 0.65 and 0.70, since this range produces the best balance among the four evaluation metrics considered.

4.2 Model Results

After estimating the final model, the resulting confusion matrix was obtained:

	Predicted Risk	
Actual Risk	Bad	Good
Bad	32	14
Good	22	105

The final accuracy reached 0.79, which represents a good although not exceptional result. Accuracy is useful for obtaining a general overview of model performance; however, it may be misleading in imbalanced datasets. Since approximately 70% of customers belong to the good class, even a naïve model predicting all customers as good would achieve an accuracy close to 0.70 without actually discriminating between the two classes.

The main evaluation metric considered in this analysis is the F1-score, defined as the harmonic mean between Precision and Recall (Sensitivity). Precision measures how reliable the positive predictions of the model are, while Recall measures the model's ability to correctly identify positive customers. Balancing these two metrics improves the reliability of positive classifications while reducing the risk of classifying risky customers as reliable.

The final F1-score for good customers is equal to 0.85, indicating that the model performs very well in identifying reliable customers. Greater difficulties emerge when identifying risky customers, where the F1-score decreases to 0.63. This result was expected due to the class imbalance in the dataset, and focusing more directly on risky customers could represent an important future improvement for the analysis.

4.3 Interpretation of the Coefficients

Once the final model was estimated, the statistical significance of the variables was evaluated. In total, 11 variables out of the original 49 were statistically significant (all categorical variables were transformed into dummy variables, explaining the increase from 20 to 49 variables). The significant variables were: duration, status_... >= 200 DM, status_no checking account, credit_history_critical account, credit_history_delay in paying off in the past, credit_history_existing credits paid back duly till now, purpose_retraining, savings_... >= 1000 DM, personal_status_sex_male : single, other_debtors_guarantor, and foreign_worker_yes.

Variables associated with higher-risk customers have negative coefficients because risky customers were encoded as class 0, meaning negative coefficients shift predicted probabilities toward 0. The variables with negative coefficients were duration, purpose_retraining, and foreign_worker_yes. These results suggest that riskier customers tend to request longer loans, while additional risk may also be associated with foreign worker status and loans requested for retraining or educational purposes.

These characteristics may describe customers experiencing temporary employment instability or weaker income conditions. However, this interpretation should be treated with caution. The purpose of this work is not to draw strong social conclusions, and these results may reflect historical and demographic biases present in the original dataset rather than genuine causal relationships.

More broadly, these findings highlight one of the major challenges in credit risk modeling: balancing predictive performance with fairness and ethical considerations.

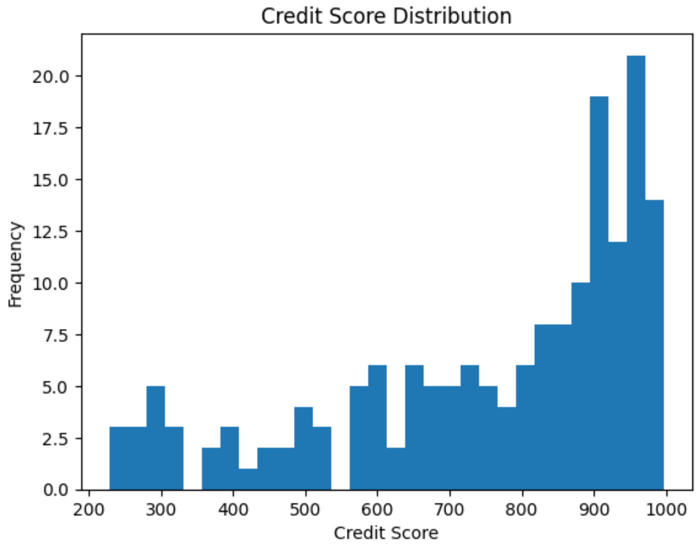
5. Credit Scoring and Risk Segmentation

To improve interpretability, the final predicted probabilities generated by the model were transformed into a credit score ranging from 0 to 1000. Higher scores indicate more reliable customers, while lower scores are associated with riskier applicants.

Risk segments were defined using score thresholds in order to create a simple and interpretable customer classification framework. The thresholds were selected to obtain a meaningful separation between highly reliable customers, intermediate-risk profiles, and potentially risky applicants.

Customers were divided into three categories: Low Risk, Medium Risk, and High Risk. Customers belonging to the Low Risk category are subject to fewer controls and faster approval procedures. Medium Risk customers require additional checks, monitoring activities, and, when necessary, manual review. Finally, High Risk customers may face rejection or more in-depth verification procedures.

Graph 3 illustrates the distribution of credit scores:

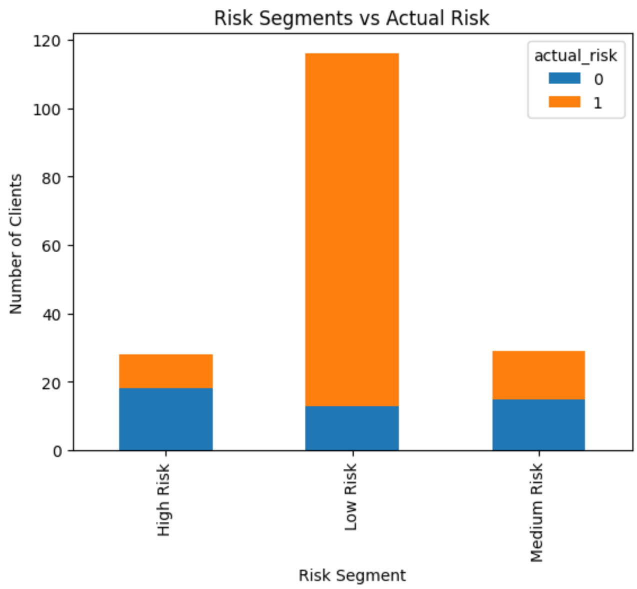


Graph 3- Credit risk distribution

The thresholds used for customer classification were defined as follows:

- Low Risk for scores above 700
- Medium Risk for scores between 500 and 700
- High Risk for scores below 500

The resulting segmentation showed coherent patterns, with the High Risk group containing a substantially larger proportion of risky customers, while the Low Risk segment was dominated by reliable applicants.



Graph 4- Segmentation results compared to actual risk

The scoring and segmentation process transformed raw model predictions into a more operational and business-oriented framework, improving interpretability and supporting potential decision-making applications.

6. Conclusions

The objective of this project was to develop a credit risk model capable of estimating customer creditworthiness using machine learning techniques and business intelligence tools. Through the use of the German Credit Dataset, it was possible to build a complete workflow including data preparation, preprocessing, statistical modeling, performance evaluation, and result visualization.

A Logistic Regression model was selected for the modeling phase due to its interpretability and its widespread use within the banking and financial sectors. The model achieved satisfactory performance, reaching a ROC-AUC of approximately 0.80 and demonstrating a good ability to distinguish between reliable and risky customers.

The evaluation metrics highlighted that the model was particularly effective in identifying reliable customers, while still maintaining a reasonable capability to detect riskier profiles. Furthermore, the threshold tuning phase allowed the selection of a decision threshold consistent with business objectives and with the trade-off between precision and recall.

Subsequently, the probabilities generated by the model were transformed into an interpretable credit scoring system, combined with a risk segmentation framework consisting of three categories: Low Risk, Medium Risk, and High Risk. This transformation made the results easier to interpret from an operational perspective and more aligned with real-world evaluation systems used in the financial sector.

Overall, the project represents a concrete example of the integration between data science, statistical analysis, and business intelligence applied to credit risk management.

7. Limitations and Future Improvements

Despite the satisfactory results obtained, the project presents several limitations that should be considered when interpreting the model’s performance.

The main limitation concerns the dataset used. The German Credit Dataset contains only 1000 observations and was created during the 1990s, making it relatively small and partially outdated compared to modern economic and financial environments. Furthermore, some variables may reflect historical characteristics or socioeconomic biases that are no longer fully representative of current reality. Given the relatively small size of the dataset, a cross-validation approach could provide more robust and stable estimates of model performance compared to a single train-test split. Techniques such as k-fold cross-validation would allow the model to be evaluated across multiple data partitions, reducing sensitivity to random sampling.

Another limitation is the class imbalance, with reliable customers representing the majority of observations compared to risky customers. This characteristic makes the identification of default cases more difficult and may negatively influence some evaluation metrics.

From a modeling perspective, Logistic Regression provides high interpretability but may not be able to capture non-linear relationships or complex interactions among variables. More advanced models such as Random Forest, XGBoost, or Gradient Boosting could potentially improve predictive performance. Outlier detection could also be improved using influence-based methods such as Cook's Distance, which are particularly suitable for regression models and allow the identification of highly influential observations.

Feature engineering could also be further improved through the creation of additional derived variables, non-linear transformations, or more sophisticated techniques for handling outliers and categorical variables.

The current model was optimized using the original target encoding, where "good" customers were represented by class 1. However, in real-world credit risk applications, the primary business objective is often the identification of risky customers and potential defaults. Future improvements could therefore include redefining the target variable in order to directly optimize the detection of bad customers, potentially improving recall and F1-score for the risky class. This approach would align the modeling strategy more closely with the economic costs associated with misclassifying high-risk applicants.

Moreover, fairness and ethical considerations represent particularly important aspects in credit scoring systems. Some variables included in the dataset, such as foreign_worker, may introduce undesirable biases into automated decision-making processes. In real-world applications, these issues require careful attention from both regulatory and ethical perspectives.

As a possible future development, the project could be extended by introducing ensemble models, more advanced cross-validation techniques, automatic hyperparameter optimization, and explainable AI systems in order to further improve interpretability and prediction reliability.